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Roll No

MI-4003-CBGS

B.E. IV Semester

Examination, June 2020

Choice Based Grading System (CBGS)

Underground Coal Mining

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

1. a) What is sublevel caving? Explain in brief.
b) Explain ventilation arrangements for a longwall retreating face with suitable layout.
2. Explain various types methods of goaf stowing.
3. What are the different types of mining methods? Explain in detail.
4. Discuss the various provisions of coal mines regulations related with mining under water bearing strata and mining of seams prone to spontaneous heating.
5. a) Compare stowing with caving method by mentioning their applicability conditions, merits and demerits.
b) Explain U/G coal gasification in brief.
6. Write short note on :
 - a) Cycle operations
 - b) Room and pillar method
 - c) Long wall method

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7. What is meant by line of extraction and its significance in depillaring operation? Explain depillaring of panes using diagonal line of extraction.

OR

Discuss the various arrangements required for depillaring of a panel developed by Board and Pillar method as per CMR.

8. a) Discuss the various factors affecting the choice of coal mining methods.
b) Explain in situ theory of coal formation. What are the various types of coal measuring rocks?
